SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1710**

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5 **Time Duration: 1 Hour**
Total Marks:30

Annexure A

Resolution Algorithm Implementation

Code of Conduct:

I SHAKTHI 192312538 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Shakthi

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Student 19: M SHAKTHI | Register Number: 192312538

Question 1 - Account Verification

Knowledge Base

1. If a user uploads a valid ID, then the account is verified
2. The user uploaded a valid ID

Goal

Prove that **the account is verified**, using the Resolution Algorithm

Tasks

1. Represent the statements in propositional logic
2. Convert all statements into CNF. Negate the goal
3. Apply the Resolution Algorithm step by step
4. Show the Empty Clause (Box) if the goal is proved

Question 2 - Premium Subscription Access

Knowledge Base

1. If a customer pays the subscription fee, then the customer gets premium access
2. Anitha paid the subscription fee

Goal

Prove that **Anitha gets premium access**, using the Resolution Algorithm

Tasks

1. Represent the statements in propositional logic
2. Convert all statements into CNF. Negate the goal
3. Apply the Resolution Algorithm step by step
4. Show the Empty Clause (Box) if the goal is proved

Question 3 - Smart Device Connectivity

Knowledge Base

1. If a device is connected to WiFi, then the device can access the internet
2. The smart TV is connected to WiFi

Goal

Prove that **the smart TV can access the internet**, using the Resolution Algorithm

Tasks

1. Represent the statements in propositional logic
2. Convert all statements into CNF. Negate the goal
3. Apply the Resolution Algorithm step by step
4. Show the Empty Clause (Box) if the goal is proved

Question 4 - Machine Auto Shutdown

Knowledge Base

1. If a machine overheats, then the machine shuts down automatically
2. The machine overheated

Goal

Prove that **the machine shuts down automatically**, using the Resolution Algorithm

Tasks

1. Represent the statements in propositional logic
2. Convert all statements into CNF. Negate the goal
3. Apply the Resolution Algorithm step by step
4. Show the Empty Clause (Box) if the goal is proved

Question 5 - Access Control System

Knowledge Base

1. If a user enters the correct password, then the user is authenticated
2. If a user is authenticated, then the user is granted access
3. The user entered the correct password

Goal

Prove that **the user is granted access**, using the Resolution Algorithm

Tasks

1. Represent the knowledge base in propositional logic
2. Convert all implications into CNF
3. Negate the goal
4. Apply the Resolution Algorithm to derive the Empty Clause (Box)
5. Explain each resolution step and write the final conclusion

Account verification

Knowledge Base

1. If a user uploads a valid ID, then the account is verified

2. The user uploaded a valid ID.

Propositional logic.

let:

- V = user uploads a valid ID
- A = Account is verified

Statements:

- $V \to A$
- V

convert to CNF: $V \to A = \neg V \vee A$

So the clauses are:

- $C_1: \neg V \vee A$
- $C_2: V$

Negate the goal: Goal: A

Resolution:
- $C_1: V \vee A$
- $C_2: V$

Resolve C_1 and C_2: A

Since the empty clause is obtained, the account is verified.

2. Premium subscription Access

Let:

- P = Anitha pays subscription fee, then the customer gets premium access

- Anitha paid the subscription fee

Propositional logic

let :

- P = Anitha pays the subscription fee
- A = Anitha gets premium access

Statements :

- $P \to A$
- P

convert to CNF

$P \to A = P \vee A$

clauses :

- C_1 : $-P \vee A$
- C_2 : P

Negate the goal

Goal : A

Negated goal : $-A$

Resolution :

Resolve :

- $\neg P \vee A$
- P

Result : A

Now Resolve :
- A
- $\neg A$

Therefore Anitha gets premium access

Smart Device Connectivity

Knowledge Base:

- If a device is connected to wi-Fi, then the device can access the internet

- The Smart TV is connected to wi-fi

Propositional logic:

Let:

- W = smart TV is connected to wi-fi
- I = smart TV can access the internet

Statements:

- $W \rightarrow I$ • W

Convert to CNF

$W \rightarrow I = \neg W \vee I$

clauses: • $C_1: \neg W \vee I$

$C_2: W$

Negate the goal: Goal: I, Negated goal: $\neg I$

Resolution:

- $W \vee I$
- W

Result:

I

Therefore, the Smart TV can access the internet

4. Machine auto shutdown

Knowledge Base

1. If a machine overheats, then the machine shuts down automatically

2. The machine overheated.

Propositional logic

let:

- O = Machine over heats
- S = Machine shuts down automatically

Statements: • $O \rightarrow S$
• O

convert to CNF

$O \rightarrow S = OVS$

Clauses:

• C_1: OVS • C_2: O

Negate the goal

Goal: S

Negated goal: $\neg S$

Resolution

Resolve:

- OVS
- O

Result:

S

Therefore, the machine shuts down automattically

Access control system:

Knowledge Base:

1. If a user enters the correct password, then the user is authenticated

2. If a user is authenticated, then the user is granted access

3. The user entered the correct password

Propositional logic:

Let:

- P = user entered the correct password
- A = user is authenticated
- G = user is granted access

Statements:

$\Rightarrow P \rightarrow A$

$\Rightarrow A \rightarrow G$

$\Rightarrow P$

convert to CNF

$P \rightarrow A$:

$\neg P \vee A$

$A \to G$:

$\neg A \lor G$

Therefore, clauses are:

$\Rightarrow C_1 : \neg P \lor A$

$\Rightarrow C_2 : \neg A \lor G$

$\Rightarrow C_3 : P$

Negate the Goal

Goal : G

$\neg G$

Apply Resolution:

Resolve C_1 and C_3:

$\Rightarrow C_1 : \neg P \lor A$

$\Rightarrow C_3 : P$

Therefore:

A

$\Rightarrow A$

$\Rightarrow C_2 : \neg A \lor G$

Now Resolve:

$\Rightarrow G$

$\Rightarrow \neg G$

Since the empty clause is obtained, the negation of the goal leads to a contradiction.

RUBRICS FOR CALCULATION

Numerical Problems						
Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**	**Marks**
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding, misses key elements	Misinterprets or unable to understand problem	19
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method	19
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps, multiple errors	No clear process	24
Accuracy of Calculation	20%	All calculations accurate, correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations	19
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation	14

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